Who Sells Before the Fall? Investor-Type Order Flow and Tail Risk in Korean Equities

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October 2026

Preliminary draft — comments welcome

Abstract

Daily trading by foreign, institutional, and individual investors is published for every Korean stock after the close, and market participants routinely read it as a signal. Using all KOSPI and KOSDAQ stocks over the two years to September 2026 (466 signal days, about 900,000 stock-days), I find that this signal contains almost no information about average returns that survives trading costs, but it does contain robust information about the left tail. A stock that foreign investors sell heavily on one day—relative to its market capitalization—is about 1 percentage point more likely to underperform the market by 20% or more over the next 20 trading days, against a base rate of about 3%. The effect survives controls for size, value, momentum, short-term reversal, same-day and next-day price moves, volatility, beta, illiquidity, turnover, and past skewness, is present in both exchanges, and appears in all of 200 random halves of the cross-section. Foreign selling is riskier than foreign buying, and individual buying is riskier than individual selling, consistent with informed sellers trading against attention-driven buyers; this asymmetry, however, weakens markedly in the second year of the sample. Institutional flows carry little tail information. The hypotheses, variables, and tests were fixed in a written analysis plan before the confirmatory estimates were produced, and the plan specifies a forward test on data that did not exist when it was written.

Keywords: investor flows, foreign investors, individual investors, crash risk, tail risk, Korea
JEL classification: G11, G12, G14, G15

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1 Introduction

In Korea, the exchange reports for every listed stock how much foreign investors, domestic institutions, and individuals bought and sold that day. The data are released after the close, reprinted by every brokerage terminal and financial news outlet, and widely interpreted as a guide to what "smart money" is doing. A large literature asks whether these flows predict *average* returns. This paper asks a different question, closer to how the numbers are often used in practice: do they tell an investor which stocks are likely to fall sharply?

The answer is yes for one specific signal. When foreign investors sell a stock heavily on a given day—heavily relative to its market capitalization, so that the signal is not just a size ranking—the probability that the stock lags the market by at least 20% over the following month rises from about 3.1% to about 4.1%, a relative increase of roughly one third (Table 3). Heavy buying by individuals has a similar effect. These estimates come from daily cross-sectional regressions that control for eleven characteristics known to predict volatility and crashes: size, book-to-market, momentum, short-term reversal, the signal-day and entry-day returns, recent volatility, beta, Amihud illiquidity, turnover, and past negative skewness. They assume a position taken at the close of the day *after* the flows are observed, so the information is actually available.

At the same time, the flows are nearly useless for predicting average returns. Moving from the most-sold to the most-bought stock by foreign investors predicts a next-day market-adjusted return difference of about 0.05%, which does not survive a multiple-testing correction, and nothing at five or twenty days (Table 2). A size-neutral long–short "flow factor" turns over more than twice its capital every day; its before-cost alpha against market, size, value, momentum, and reversal factors is small and statistically insignificant, and after a modest one-way cost of 0.125% it loses about 0.26% per day. The contrast is the main point of the paper: the same public signal that cannot be traded profitably for its mean carries economically meaningful information about the left tail.

The tail effect is directional. Foreign selling is riskier than foreign buying by 0.72 percentage points, and individual buying is riskier than individual selling by 0.97 percentage points (Table 4). Because every trade has two sides, heavy foreign selling and heavy individual buying often coincide; when all three groups' signals enter the same regression, about half of the foreign-selling effect remains and most of the individual-buying effect remains. The natural reading is a familiar one in the literature on investor types: sellers with better information or faster access to it trade against buyers whose purchases are driven by attention or by a contrarian habit, and when the information is bad the price adjustment arrives as a large drop rather than a gradual drift. Institutional flows, by contrast, are weak predictors in either direction.

The results are robust to the timing of entry (open or close of the next day), the threshold that defines an extreme flow (5%, 10%, 20%, or a heavy-trading variant), the definition of the tail event (an end-of-horizon or a path minimum), the exchange, the size of the stock, the state of the market, the treatment of missing data, and the estimation method (Table 5, Figure 1). The foreign-selling coefficient is almost identical in the first and second years of the sample, and it is positive and significant in both halves of every one of 200 random splits of the cross-section (Table 6). Two

qualifications are important. First, the information is in *one-day* selling shocks: flows cumulated over 5 or 20 days do not predict tail events, and persistent 20-day foreign selling is, if anything, associated with *lower* tail risk. Second, the asymmetry between selling and buying weakens sharply in the second year (Figure 2); the level effect of foreign selling does not.

The paper makes three contributions. First, it moves the question about investor-type flows from the mean to the tail. Prior work on Korea and on other markets with investor-type data has studied whether foreign or individual trading predicts returns or destabilizes prices (e.g., Choe et al., 1999, 2005; Richards, 2005; Grinblatt and Keloharju, 2000; Kaniel et al., 2008); the crash-risk literature has studied firm-level predictors such as opacity, trading volume, and past returns (e.g., Chen et al., 2001; Jin and Myers, 2006; Hutton et al., 2009). This paper links the two and shows that the public flow signal is a crash predictor even though it is not a return predictor. Second, it quantifies the predictive content in a form that risk managers can use directly—a change in the probability of a -20% relative loss—and shows that this quantity, unlike the mean, is stable across specifications. Third, it is conducted under a written analysis plan that fixed hypotheses, variable definitions, thresholds, controls, inference, multiple-testing rules, and a list of robustness checks before the confirmatory estimates were produced. The plan also discloses which results had been seen in earlier exploration, and it schedules a test on future data. Given the short sample, I regard this transparency as essential rather than decorative (Harvey et al., 2016; Nosek et al., 2018).

The sample is short by design. It covers the two most recent years because the question is about the current market structure, in which order flow is published and consumed in near real time and regulatory conditions differ from earlier decades. The cost of this choice is statistical power and the risk that the period is unusual: the KOSPI more than doubled while the KOSDAQ was roughly flat. I address the first concern by relying on the very large cross-section, by using inference that accounts for overlapping horizons, and by reporting split-sample replications. I address the second by showing results for rising and falling markets and for each exchange separately, and by committing to an out-of-sample evaluation on data after October 2026. Section 8 discusses what the results do and do not establish.

2 Related literature and hypotheses

2.1 Investor types and returns

A large body of work uses account-level or investor-type data to ask who trades better. In Korea, Choe et al. (1999) find little evidence that foreign investors destabilized prices during the 1997 crisis, and Choe et al. (2005) find that domestic investors trade at better prices than foreign investors in the same stocks, suggesting a domestic information advantage at short horizons. Kim and Wei (2002) document foreign investors' positive-feedback trading and herding in Korea. For a broader set of Asian markets, Richards (2005) shows that foreign flows have a sizable contemporaneous price impact. In Finland, Grinblatt and Keloharju (2000) report that foreign investors outperform and households tend to be contrarian. In the United States, Kaniel et al. (2008) show that individuals

are contrarians whose net buying is followed by positive returns, consistent with liquidity provision, while Barber and Odean (2008) show that individual purchases are concentrated in attention-grabbing stocks. Griffin et al. (2003) relate daily institutional and individual trading to past and future returns.

The present paper differs from this work in two ways. It studies the *publicly disseminated* daily investor-type signal rather than private account data, and it focuses on the probability of large relative losses rather than on mean returns. Its average-return results (Section 5) are consistent with the view that whatever mean predictability exists in this public signal is small and short-lived.

2.2 Crash risk

Chen et al. (2001) introduce the negative coefficient of skewness (NCSKEW) and the down-to-up volatility ratio (DUVOL) of firm-specific returns and show that conditional skewness is predictable from trading volume and past returns. Jin and Myers (2006) and Hutton et al. (2009) argue that firms that withhold or obscure bad news are more crash-prone because the news is eventually released at once. That mechanism maps naturally onto the investor-type setting: if a subset of investors learns bad news early and sells, the selling is a noisy public footprint of news that has not yet been fully reflected in the price.

2.3 Hypotheses

The analysis plan specifies the following hypotheses. Let β_S^j (β_B^j) be the change in the probability of a large 20-day underperformance associated with extreme selling (buying) by investor group $j \in \{F, I, R\}$ (foreign, institutional, individual).

- H1 $\beta_S^F > 0$: heavy foreign selling predicts tail events after controls.
- H2 $\beta_S^F - \beta_B^F > 0$: foreign selling is riskier than foreign buying.
- H3 $|\beta_S^I - \beta_B^I| < 0.5$ percentage points: institutional selling and buying carry the same tail risk.
- H4 $\beta_B^R - \beta_S^R > 0$: individual buying is riskier than individual selling.
- H5–H6 (secondary) extreme flows of any type raise tail risk; the foreign-selling effect survives when all groups' signals are included together.
- H7–H8 (secondary) foreign net buying predicts positive one- to five-day returns, but a tradable flow factor has no alpha after costs.

H1, H2, and H4 form the primary family. H3 is an equivalence hypothesis, motivated by exploratory evidence that institutional selling, often driven by rebalancing and fund flows, is uninformative.

3 Data

3.1 Sources and sample

All data are from the Korea Exchange (KRX). For each trading day and each listed stock I obtain the buy value, sell value, and net buy value of three investor groups—foreign investors, institutions (all domestic institutions combined), and individuals—together with prices, trading volume, trading value, the official daily return, market capitalization, and book value per share. The risk-free rate is the 91-day certificate of deposit (CD) rate.

The signal sample consists of 466 trading days from 22 October 2024 to 17 September 2026; the last date is the last day for which a full 20-day outcome window is observed. Prices from September 2023 to October 2024 are used only to compute lagged control variables (momentum and beta) and do not enter the outcome sample. On each day the universe includes common stocks listed on the KOSPI or KOSDAQ with positive trading volume and market capitalization of at least KRW 50 billion, excluding special-purpose acquisition companies and real-estate investment trusts. This leaves about 1,900 stocks per day and 898,838 stock-days. Stocks that are delisted during a holding period are kept: their return is measured to the last traded price and set to zero afterwards, so the sample is free of survivorship bias.

3.2 Flow signals

The net-buying intensity of investor group j in stock i on day d is

$$x_{i,d}^{j} = \frac{N_{i,d}^{j}}{M_{i,d}}, \tag{1}$$

where $N_{i,d}^{j}$ is net purchase value (buy minus sell value, in won) and $M_{i,d}$ is the stock's market capitalization. Scaling by market capitalization prevents the signal from simply ranking stocks by size. Each day, within each exchange, I flag the bottom decile of $x_{i,d}^{j}$ as extreme selling ($S_{i,d}^{j} = 1$) and the top decile as extreme buying ($B_{i,d}^{j} = 1$); the middle 80% is the comparison group.

3.3 Outcomes

Because the flows are published after the close, a position based on day-d flows can be taken at the close of day $d+1$ at the earliest. The h-day holding return is $R_{i,d}^{h} = \prod_{t=d+2}^{d+1+h}(1 + r_{i,t}) - 1$, and the market-adjusted return $e_{i,d}^{h}$ subtracts the equal-weighted average R^{h} of all universe stocks on the same exchange on the same day. The primary tail outcome is

$$Y_{i,d} = \mathbf{1}\{e_{i,d}^{20} \leq -0.20\}, \tag{2}$$

an underperformance of the market by 20 percentage points or more over the 20 trading days after entry. Its unconditional frequency is 4.5%, and about 3.1% for stocks without an extreme

signal. As secondary outcomes I use NCSKEW and DUVOL of firm-specific log returns $w_{i,t} = \ln(1 + r_{i,t} - \hat{\beta}_{i,d} r_{m,t})$ over the holding window, where $\hat{\beta}_{i,d}$ is estimated from data up to day d and $r_{m,t}$ is the value-weighted return of the stock's exchange (Chen et al., 2001).

3.4 Controls and factors

The eleven controls $Z_{i,d}$ are all known at the time of entry: log market capitalization, log book-to-market, momentum (cumulative return from month $t-12$ to $t-1$), short-term reversal (prior 20-day return), the signal-day return $r_{i,d}$, the entry-day return $r_{i,d+1}$, 20-day return volatility, 250-day beta, Amihud illiquidity (Amihud, 2002), 20-day turnover, and past 60-day NCSKEW. Controls are winsorized daily at the 1st and 99th percentiles. Table 1 reports summary statistics, and Appendix A gives exact definitions.

For the return analysis I construct Korean versions of the market, size, value (Fama and French, 1993), momentum (Carhart, 1997), and short-term reversal (Jegadeesh, 1990) factors from the same universe (Appendix Table 8). My value-weighted exchange indices have daily return correlations of 1.0000 (KOSPI) and 0.9996 (KOSDAQ) with the official KRX indices.

Table 1: Summary statistics

Variable	Obs.	Mean	Std. dev.	P25	Median	P75
Tail event Y ($e^{20} \leq -20\%$)	898,838	0.0449	0.2072	0.0000	0.0000	0.0000
Path minimum $\leq -20\%$	898,838	0.1134	0.3171	0.0000	0.0000	0.0000
20-day excess return e^{20}	898,838	-0.0000	0.1742	-0.0836	-0.0233	0.0476
NCSKEW (20 days)	874,314	-0.1541	1.2821	-1.0342	-0.1114	0.7798
DUVOL (20 days)	874,309	-0.1050	1.3529	-0.9883	-0.1005	0.7971
Foreign net buying x^F (bp of cap)	898,838	-0.0971	23.7199	-4.2110	-0.0465	4.3370
Institutional net buying x^I (bp)	898,838	-0.2274	19.5071	-0.4738	0.0000	0.3759
Individual net buying x^R (bp)	898,838	0.5095	33.8946	-4.7432	-0.0002	4.4587
Log market cap	898,838	26.2698	1.3877	25.2394	25.8928	26.8822
Log book-to-market	886,221	-0.1845	1.0532	-0.8215	-0.0443	0.6093
Momentum (12–1)	856,741	0.1523	0.6814	-0.2135	-0.0199	0.2762
Short-term reversal (20 days)	895,476	0.0135	0.1695	-0.0753	-0.0065	0.0721
Signal-day return	898,838	0.0005	0.0358	-0.0165	-0.0011	0.0136
Entry-day return	898,838	0.0000	0.0353	-0.0167	-0.0012	0.0133
Volatility (20 days)	894,576	0.0333	0.0214	0.0179	0.0277	0.0424
Beta (250 days)	877,653	0.7529	0.3595	0.4737	0.7260	1.0046
Amihud illiquidity	894,576	0.0050	0.0081	0.0007	0.0023	0.0057
Turnover (20 days)	894,576	0.0156	0.0329	0.0023	0.0049	0.0125
Past NCSKEW (60 days)	875,522	-0.5643	1.2654	-1.3050	-0.4694	0.2614

Notes: The sample covers 466 signal days (22 October 2024 to 17 September 2026) and 898,838 stock-days. Returns are measured from the close of day $d+1$. Control variables are winsorized at the 1st and 99th percentiles each day. Net buying is the investor group's net purchase value on day d divided by the stock's market capitalization, in basis points. Variable definitions are in Appendix A.

4 Methodology

4.1 Tail-risk regressions

On each signal day d and for each investor group j I estimate the cross-sectional linear probability model

$$Y_{i,d} = a_d + \beta^j_{S,d} S^j_{i,d} + \beta^j_{B,d} B^j_{i,d} + \delta_d \,\mathrm{KOSPI}_i + \gamma'_d Z_{i,d} + \varepsilon_{i,d}. \tag{3}$$

The coefficients $\beta^j_{S,d}$ and $\beta^j_{B,d}$ measure, in probability units, how much more likely an extreme-flow stock is to suffer a tail event than an otherwise similar stock in the middle of the flow distribution. Estimating the regression day by day absorbs any market-wide shock into the intercept. Following Fama and MacBeth (1973), the reported coefficient is the time-series average $\bar{\beta} = T^{-1}\sum_d \beta_d$. Because consecutive 20-day outcome windows overlap by 19 days, the daily coefficients are serially correlated; I therefore use Newey–West standard errors (Newey and West, 1987) with $L = \max(h - 1, 5)$ lags, i.e., 19 lags for the 20-day outcome. Differences such as $\beta^F_S - \beta^F_B$ are computed day by day and then averaged. A "joint" version of (3) includes the extreme-flow indicators of all three groups simultaneously.

4.2 Average-return regressions and flow factors

To study average returns I replace Y with $e^h_{i,d}$ ($h = 1, 5, 20$) and the indicators with the within-exchange percentile rank of net buying minus one half, $q^j_{i,d} \in [-0.5, 0.5]$, so that the slope is the return difference between the most-bought and most-sold stock. I also build, for each group, a size-neutral long–short portfolio $\mathrm{FLOW}^j = \frac{1}{2}(R_{SH} + R_{BH}) - \frac{1}{2}(R_{SL} + R_{BL})$ from a 2×3 sort on market capitalization and x^j, formed with day-d flows at the close of $d + 1$ and value weighted. I regress its daily returns on MKT, SMB, HML, MOM, and STR, both before costs and after deducting daily turnover times a one-way cost of 0.125% (half of a 0.25% round-trip cost that includes the transaction tax).

4.3 Inference and the analysis plan

The hypotheses, variable definitions, thresholds, controls, estimators, and decision rules were written down in an analysis plan dated 27 October 2026, after exploratory work on the same two years of data but before any of the estimates reported in Sections 5–7 were produced. The primary family (H1, H2, H4) is tested one-sided with Holm's step-down correction at 5% (Holm, 1979). H3 is tested with two one-sided tests (Schuirmann, 1987) against a margin of ± 0.5 percentage points, which is half of the foreign-selling asymmetry seen in exploration. Secondary hypotheses are reported with Bonferroni-adjusted thresholds. The plan also lists all robustness checks reported in Section 7 and requires that every one be reported. Section 9 lists all deviations and disclosures.

5 Flows and average returns

Panel A of Table 2 shows that investor flows have little predictive power for average returns once standard characteristics are controlled for. The only positive estimate with a t-statistic above two is foreign flow at a one-day horizon: moving from the most-sold to the most-bought stock raises the next-day market-adjusted return by 0.050% ($t = 2.03$). This is significant at the 5% level before adjustment but not after the Bonferroni correction for the secondary family, and it disappears at five and twenty days. Institutional and individual flows predict nothing.

Panel B translates the signal into tradable portfolios. Rebalanced daily, the foreign flow factor earns a before-cost alpha of 0.069% per day ($t = 1.63$), but it trades 2.6 times its long and short capital every day. Its break-even one-way cost is 0.026%, well below realistic costs, and after a 0.125% one-way cost its alpha is -0.26% per day. Rebalancing every five days reduces turnover to about 0.5 and leaves a before-cost alpha of 0.075% ($t = 1.83$) with a break-even cost of 0.137%, close to the assumed cost; the after-cost alpha is essentially zero. Institutional and individual factors have no alpha before costs. The factor loadings are intuitive: foreign flows tilt toward small and value stocks, and the individual flow factor loads positively on short-term reversal, consistent with individuals buying recent losers.

One pre-specified hypothesis needs comment. H8 was phrased as "the after-cost alpha is not different from zero" (two-sided). The after-cost alphas of the daily factors are significantly *negative*, so H8 is formally rejected. Substantively the result is stronger than anticipated—costs do not merely eliminate profits, they produce losses—but I report the formal outcome as planned.

6 Flows and tail risk

6.1 Main results

Table 3 reports the tail-risk regressions. Stocks that foreign investors sold most heavily on day d are 0.98 percentage points more likely to underperform the market by at least 20% over the following 20 trading days ($t = 6.30$). The base rate for stocks without an extreme signal is 3.10%, so the relative increase is about 32%. Stocks that foreign investors bought most heavily show a much smaller and insignificant increase (0.26 points, $t = 1.49$). For individuals the pattern is reversed: heavy individual buying raises the tail probability by 1.16 points ($t = 6.71$), while heavy individual selling does not (0.19 points). Neither institutional signal is significant.

Table 4 reports the pre-specified tests. All three primary hypotheses are supported after Holm's correction. The foreign selling-minus-buying difference is 0.72 points ($t = 4.04$), and the individual buying-minus-selling difference is 0.97 points ($t = 4.14$). H3 is not established: the institutional difference is 0.41 points ($t = 1.96$), and the TOST procedure cannot rule out differences larger than half a percentage point. Institutional selling is thus, if anything, slightly riskier than institutional buying, although neither institutional coefficient alone is significant. Among the secondary hypotheses, the idea that extreme flows of any kind raise tail risk (H5) holds only for foreign sell-

Table 2: Investor flows and average returns

Panel A: Fama–MacBeth regressions of market-adjusted returns on the net-buying rank

Investor group	$h = 1$	$h = 5$	$h = 20$
Foreign	0.050	0.077	0.093
	(2.03)	(1.28)	(0.65)
Institutional	-0.013	-0.069	-0.241
	(-0.60)	(-1.02)	(-1.44)
Individual	-0.042	-0.051	0.129
	(-1.63)	(-0.77)	(0.66)

Panel B: Size-neutral flow factors (daily returns, %)

Investor group	Rebalancing	Mean	α before costs	α after costs	Turnover	Break-even cost
Foreign	Daily	0.040 (0.85)	0.069 (1.63)	-0.259 (-6.11)	2.62	0.026
Foreign	5-day	0.039 (0.90)	0.075 (1.83)	0.007 (0.16)	0.54	0.137
Institutional	Daily	-0.001 (-0.02)	-0.005 (-0.12)	-0.293 (-7.40)	2.30	–
Institutional	5-day	0.049 (1.29)	0.032 (0.88)	-0.031 (-0.83)	0.50	0.064
Individual	Daily	-0.025 (-0.52)	-0.049 (-1.06)	-0.380 (-8.17)	2.64	–
Individual	5-day	-0.055 (-1.27)	-0.081 (-1.88)	-0.150 (-3.47)	0.55	–

Notes: Panel A reports time-series averages (in %) of daily cross-sectional slopes from $e^h_{i,d} = a_d + b_d q^j_{i,d} + \delta\,\mathrm{KOSPI}_i + c' Z_{i,d} + u_{i,d}$, where q^j is the within-market percentile rank of net buying minus 0.5, so b is the return difference between the most-bought and the most-sold stock. Newey–West t-statistics with $\max(h-1, 5)$ lags are in parentheses. Panel B reports $\mathrm{FLOW}^j = \frac{1}{2}(R_{SH} + R_{BH}) - \frac{1}{2}(R_{SL} + R_{BL})$ from a 2×3 sort on size and net buying (30/70 breakpoints), formed with day-d flows at the close of $d+1$, value weighted. α is the intercept from regressing FLOW on MKT, SMB, HML, MOM and STR (Newey–West, 5 lags). Turnover is traded value per unit of long and short capital per day; after-cost returns deduct turnover times a one-way cost of 0.125%. Break-even cost is the one-way cost that sets the before-cost α to zero.

ing and individual buying: an exploratory finding that heavily bought stocks are also riskier is explained by the controls, in particular volatility, the signal-day return, and turnover.

Table 3: Extreme investor flows and the probability of a large 20-day underperformance

	Separate regressions				Joint regression	
Investor group	β_S	β_B	$\beta_S - \beta_B$	Base rate	β_S	β_B
Foreign	0.98	0.26	0.72	3.10	0.40	0.16
	(6.30)	(1.49)	(4.04)		(2.98)	(1.07)
Institutional	0.16	-0.26	0.41	3.64	-0.09	-0.34
	(0.53)	(-1.19)	(1.96)		(-0.33)	(-1.66)
Individual	0.19	1.16	-0.97	3.09	0.18	0.93
	(1.07)	(6.71)	(-4.14)		(1.11)	(5.81)

Notes: Each day d we estimate the cross-sectional linear probability model $Y_{i,d} = a_d + \beta_S S^j_{i,d} + \beta_B B^j_{i,d} + \delta\,\mathrm{KOSPI}_i + \gamma' Z_{i,d} + \varepsilon_{i,d}$, where $Y_{i,d} = 1$ if the stock's market-adjusted return over the 20 trading days after the close of $d+1$ is -20% or worse, S^j (B^j) flags the bottom (top) decile of investor group j's net buying scaled by market capitalization on day d within the stock's market, and Z contains the eleven controls of Table 1. Coefficients are time-series averages of the daily estimates, in percentage points. Newey–West t-statistics with 19 lags are in parentheses ($T = 466$ days). The base rate is the average daily frequency of $Y = 1$ among stocks with neither signal. The joint regression includes all six signals.

Table 4: Pre-specified hypothesis tests: tail risk

	Hypothesis	Estimate (pp)	t	p	Decision rule	Result
H1	Foreign extreme selling raises tail risk ($\beta^F_S > 0$)	0.98	6.30	<0.0001	Holm, 0.0167	Supported
H2	Foreign selling riskier than buying ($\beta^F_S - \beta^F_B > 0$)	0.72	4.04	<0.0001	Holm, 0.0500	Supported
H4	Individual buying riskier than selling ($\beta^R_B - \beta^R_S > 0$)	0.97	4.14	<0.0001	Holm, 0.0250	Supported
H3	Institutional equivalence ($\lvert\beta^I_S - \beta^I_B\rvert < 0.5$pp, TOST)	0.41	1.96	0.341	TOST, 5%	Not established
H5	Foreign selling coefficient > 0	0.98	6.30	<0.0001	Bonferroni, 0.0042	Significant
H5	Foreign buying coefficient > 0	0.26	1.49	0.0686	Bonferroni, 0.0042	Not significant
H5	Institutional selling coefficient > 0	0.16	0.53	0.2980	Bonferroni, 0.0042	Not significant
H5	Institutional buying coefficient > 0	-0.26	-1.19	0.8826	Bonferroni, 0.0042	Not significant
H5	Individual selling coefficient > 0	0.19	1.07	0.1420	Bonferroni, 0.0042	Not significant
H5	Individual buying coefficient > 0	1.16	6.71	<0.0001	Bonferroni, 0.0042	Significant
H6	Foreign extreme selling, joint model ($\beta^F_S > 0$)	0.40	2.98	0.0015	Bonferroni, 0.0042	Significant

Notes: Estimates and Newey–West t-statistics come from Table 3. p-values are one-sided under the normal approximation. The primary family (H1, H2, H4) is controlled at 5% with Holm's step-down procedure; the threshold shown is the one applied to each hypothesis. H3 is a two one-sided tests (TOST) equivalence test with margin ± 0.5 percentage points; the ordinary two-sided p-value for $\beta^I_S - \beta^I_B = 0$ is 0.049. Secondary hypotheses use a Bonferroni threshold of 0.05/12.

6.2 Who is on the other side?

Every share sold by a foreign investor is bought by someone. When the six signals enter jointly (last two columns of Table 3), the foreign-selling coefficient falls from 0.98 to 0.40 points but remains significant ($t = 2.98$), while the individual-buying coefficient falls only from 1.16 to 0.93 points. A large part of the tail-risk information therefore sits in the transfer of shares from foreign investors to individuals, and the individual side of that transfer is the stronger marker. This is what one would expect if foreign selling reflects early access to negative information and individual buying reflects attention to a falling or volatile stock (Barber and Odean, 2008).

6.3 Conditional skewness

Appendix Table 7 replaces Y with NCSKEW and DUVOL. Over 20 days, neither measure responds to the flow signals. Over 60 days the foreign selling-minus-buying difference is positive ($t \approx 2.2$ for both measures) and the individual difference has the expected sign, but these estimates rest on only seven non-overlapping windows. The contrast with Table 3 indicates that the flow signal predicts the probability of a single large relative loss rather than the shape of the daily return distribution; with 20 daily observations, skewness-based measures are also noisy.

7 Robustness and within-sample replication

7.1 Pre-specified robustness checks

Table 5 reports all 23 pre-specified variations on the baseline; Figure 1 plots the foreign-selling coefficient with 95% intervals. Entering at the next day's open instead of its close, or (as an infeasible reference) at the signal-day close, slightly strengthens all results. Defining extreme flows by the 5% or 20% tails, or by a heavy-trading variant that requires the investor group to be unusually active in the stock, leaves the conclusions unchanged, and the foreign-selling coefficient rises to 1.37 points for the most extreme 5%. The results hold in the KOSPI and the KOSDAQ separately, for stocks above KRW 100 billion, in rising and falling markets (stronger in falling markets), with missing controls imputed, with lagged book-to-market, and in pooled regressions with day fixed effects and two-way clustered standard errors (Petersen, 2009; Cameron et al., 2011). A pooled logit with day fixed effects gives the same signs; its marginal effects for the two asymmetries are smaller (0.37 and 0.30 points).

Three variations change the picture in informative ways. First, signals based on flows cumulated over 5 or 20 days do not predict tail events; persistent 20-day foreign selling is associated with a *lower* tail probability (-0.84 points, $t = -3.1$). The information is in abrupt one-day selling, not in sustained outflows, which are more likely to reflect index reweighting or slow portfolio adjustment. Second, with a -10% threshold both extreme foreign selling and extreme foreign buying raise the probability of moderate underperformance, and the selling-buying asymmetry is insignificant; the asymmetry is a feature of the deep tail. Third, over 60 days the foreign effect remains while the individual asymmetry disappears; these estimates are underpowered. Because the planned checks were designed to test whether the baseline survives rather than to discover new patterns, I label these three observations as exploratory.

7.2 Time and cross-sectional splits

Panel A of Table 6 splits the sample into its first and second years. The foreign-selling effect is virtually identical in the two years (0.99 versus 0.97 points). The two asymmetries are not: the foreign selling-minus-buying difference falls from 1.11 to 0.29 points and the individual buying-minus-selling difference from 1.54 to 0.36 points, and both declines are statistically significant

Table 5: Robustness of the tail-risk results

Specification	Foreign β_S (H1)	Foreign $\beta_S - \beta_B$ (H2)	Individual $\beta_S - \beta_B$ (H4)	Institutional $\beta_S - \beta_B$ (H3)
Baseline	0.98 (6.30)	0.72 (4.04)	-0.97 (-4.14)	0.41 (1.96)
R1: Entry at the open of $d+1$	1.15 (6.60)	0.86 (4.82)	-1.12 (-4.97)	0.49 (2.31)
R1: Entry at the close of d (infeasible, reference)	1.13 (6.71)	0.88 (5.44)	-1.08 (-5.39)	0.62 (3.05)
R2: Signal from 5-day cumulative flow	0.31 (1.33)	0.14 (0.54)	-0.73 (-2.48)	0.28 (0.91)
R2: Signal from 20-day cumulative flow	-0.84 (-3.14)	-0.63 (-1.66)	-0.40 (-1.15)	0.57 (2.14)
R3: Heavy-trading signal	1.20 (6.48)	0.83 (5.18)	-1.04 (-5.35)	0.41 (2.47)
R3: Extreme 5%	1.37 (4.98)	0.71 (2.75)	-1.09 (-4.08)	0.37 (1.13)
R3: Extreme 20%	0.56 (6.33)	0.57 (4.74)	-0.86 (-5.94)	0.33 (2.44)
R4: Threshold -10%	2.21 (6.77)	0.42 (1.54)	-0.10 (-0.22)	-0.52 (-1.02)
R4: Path minimum $\leq -20\%$	1.55 (5.36)	0.78 (3.31)	-1.22 (-3.48)	0.48 (1.19)
R5: 10-day horizon	0.47 (4.54)	0.70 (7.04)	-1.02 (-7.15)	0.49 (3.58)
R5: 60-day horizon†	1.70 (2.81)	0.67 (1.97)	-0.14 (-0.32)	-0.86 (-2.75)
R6: KOSPI only	0.75 (5.28)	0.64 (3.01)	-1.04 (-4.47)	0.85 (2.83)
R6: KOSDAQ only	1.15 (5.48)	0.78 (3.20)	-0.93 (-3.05)	0.11 (0.51)
R7: Largest size tercile	0.80 (3.73)	0.55 (2.01)	-0.45 (-1.47)	-0.03 (-0.10)
R7: Market cap $\geq$ KRW 100bn	0.77 (4.80)	0.57 (2.88)	-0.79 (-3.46)	0.34 (1.57)
R8: First year	0.99 (8.16)	1.11 (6.60)	-1.54 (-5.81)	0.54 (2.40)
R8: Second year	0.97 (3.27)	0.29 (1.16)	-0.36 (-1.22)	0.28 (0.77)
R9: Market up over prior 20 days	0.82 (4.70)	0.55 (2.44)	-0.62 (-2.44)	0.25 (1.48)
R9: Market down over prior 20 days	1.32 (5.67)	1.08 (3.13)	-1.72 (-4.58)	0.75 (1.78)
R10: Pooled LPM, day FE, two-way clustered	1.05 (7.04)	0.93 (4.60)	-1.12 (-5.00)	0.37 (1.90)
R10: Logit AME, day FE, two-way clustered‡	1.21 (13.91)	0.37 (3.67)	-0.30 (-2.72)	-0.10 (-0.74)
R11: Missing controls imputed	1.15 (6.87)	0.77 (4.40)	-1.05 (-4.51)	0.44 (2.24)
R12: Book-to-market lagged 60 days	1.01 (6.41)	0.72 (4.03)	-0.98 (-4.13)	0.41 (1.95)

Notes: Each row changes one element of the baseline specification of Table 3 (all rows were pre-specified). Entries are coefficients in percentage points with t-statistics in parentheses. H1 and H2 predict positive values, H4 predicts a negative value (individual buying riskier), and H3 predicts a value near zero. Fama–MacBeth rows use Newey–West standard errors with $\max(h-1, 5)$ lags. †Only seven non-overlapping 60-day windows; treated as underpowered. ‡Standard errors of average marginal effects are approximated by scaling coefficient standard errors with the mean logistic density and may be too small.

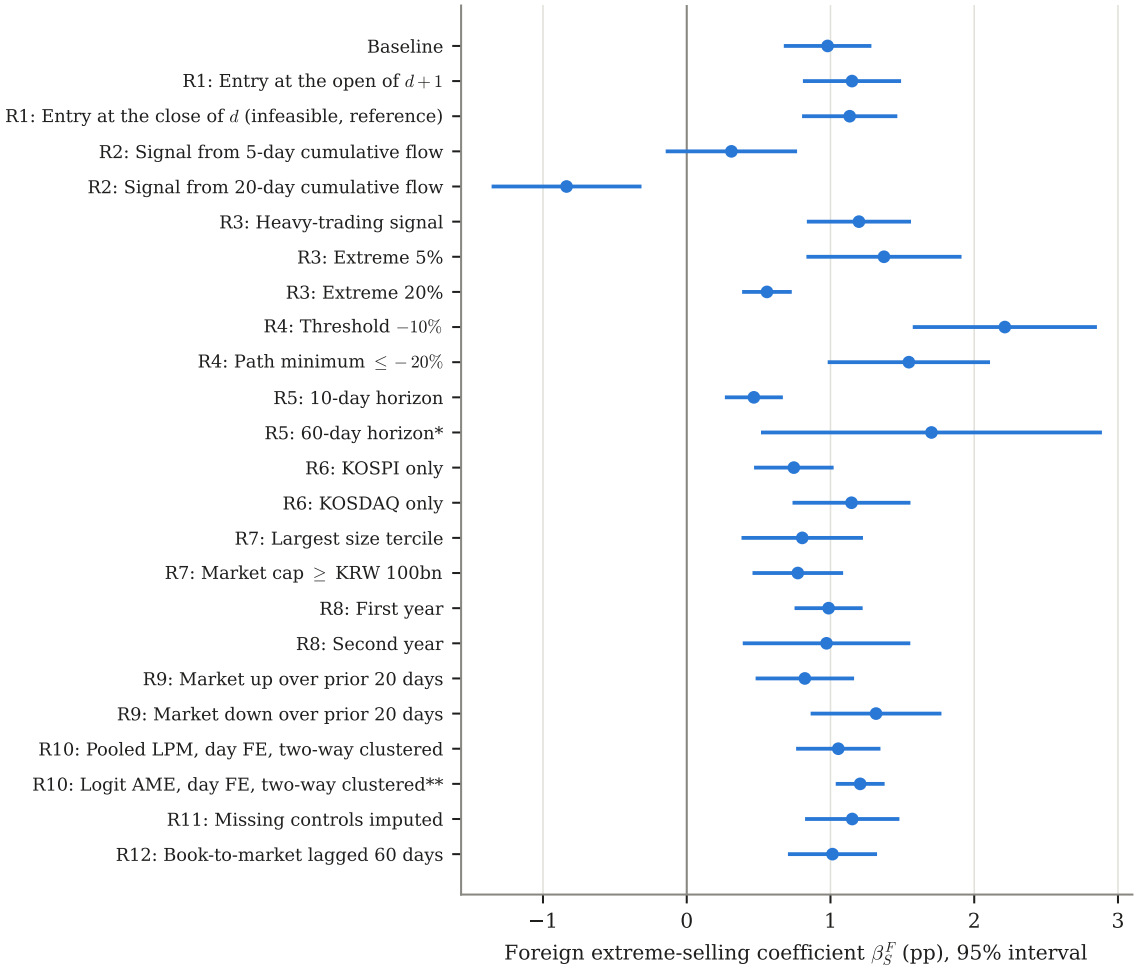


Figure 1: Foreign extreme-selling coefficient across pre-specified specifications. Points are coefficients from Table 5 (percentage points); bars are 95% intervals based on the standard errors described there. * underpowered; ** approximate standard errors.

($p = 0.007$ and $p = 0.003$). Figure 2 shows the same pattern in 60-day rolling averages of the daily coefficients. Panel B splits the cross-section into random halves 200 times. H1 is positive and significant in both halves in every split, H2 in 97.5%, and H4 in 99.5%, so no small group of stocks drives the results.

These splits use the data from which the hypotheses were derived and are not true out-of-sample tests. The analysis plan therefore specifies a forward evaluation on signal days from 23 October 2026, with a first look after 60 and a second after 120 signal days; the primary criterion is a one-sided $p < 0.05$ for H2 at the second evaluation. If the effect stays at its full-sample size the expected t-statistic after 120 days is about 2.4, but the second-year decline in Panel A makes a failure of H2 a real possibility. H1 will be reported alongside it.

Table 6: Within-sample replication: time and cross-sectional splits

Panel A: First year versus second year

	First year	Second year	z (difference)	p (two-sided)
H1: Foreign β_S	0.99 (8.16)	0.97 (3.27)	0.04	0.964
H2: Foreign $\beta_S - \beta_B$	1.11 (6.60)	0.29 (1.16)	2.69	0.007
H4: Individual $\beta_B - \beta_S$	1.54 (5.81)	0.36 (1.22)	2.98	0.003
H3: Institutional $\beta_S - \beta_B$	0.54 (2.40)	0.28 (0.77)	0.62	0.532

Panel B: Random halves of stocks, 200 splits

	Same sign in	Significant in	t-statistics (400 halves)		
	both halves (%)	both halves (%)	Min	Median	Max
H1: Foreign β_S	100.0	100.0	3.03	5.07	9.07
H2: Foreign $\beta_S - \beta_B$	100.0	97.5	1.20	3.20	5.28
H4: Individual $\beta_B - \beta_S$	100.0	99.5	1.64	3.47	5.92

Notes: Panel A re-estimates the baseline model separately for signal days 22 October 2024 to 22 October 2025 and 23 October 2025 to 17 September 2026 (coefficients in percentage points, signed so that positive values support the hypothesis; Newey–West t-statistics in parentheses). $z = (\bar{\beta}_1 - \bar{\beta}_2)/\sqrt{se_1^2 + se_2^2}$ treats the two halves as independent. Panel B splits the cross-section of stocks into two random halves (seeds 0–199) and re-estimates the baseline model in each half; "significant" means a one-sided $p < 0.05$. Both panels use data from which the hypotheses were derived and are therefore not true out-of-sample tests.

8 Discussion

8.1 Interpretation

The evidence fits a simple account. On some days a group of investors with an informational or speed advantage—in this sample, foreign investors—sells a stock aggressively relative to its size. The selling is absorbed largely by individuals. The price does not fully adjust at once, and when the underlying news is bad the remaining adjustment arrives as a large drop within weeks. Because such episodes are rare and mixed with many uninformative trades, the average return of a portfolio sorted on flows barely moves, but the frequency of large losses does. This account is consistent with theories in which bad news is accumulated and released abruptly (Jin and Myers, 2006), and with

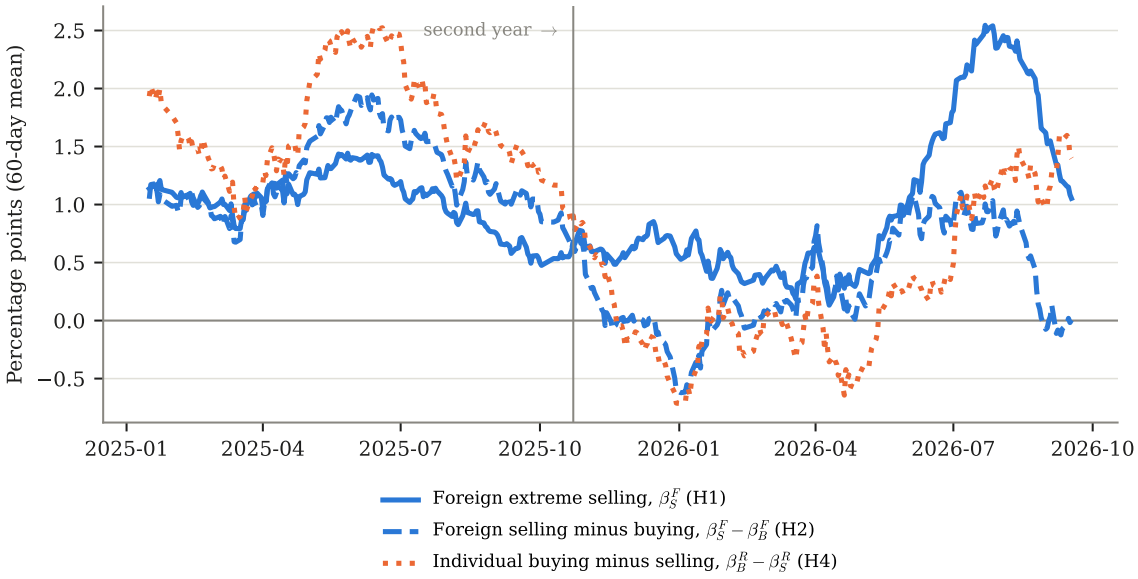


Figure 2: Rolling 60-signal-day averages of the daily cross-sectional coefficients from equation (3), in percentage points. The vertical line marks the start of the second year (23 October 2025).

evidence that individual purchases are attention driven (Barber and Odean, 2008). The finding that sustained, multi-week foreign selling is not risky suggests that the informative component is the unusual one-day trade, not the trend.

The results do not establish causality. Foreign selling could cause subsequent declines through price pressure, or foreign investors could simply anticipate declines; the regressions cannot separate the two. Nor is the signal a precise forecasting tool: even among stocks with the strongest signals, more than 95% do not experience a tail event within 20 days.

8.2 Implications for risk management

For a long-only investor, the most direct use is in position sizing and monitoring. Heavy one-day foreign selling in a held stock—especially when combined with heavy individual buying—raises the probability of a large relative loss by roughly a third, after accounting for the stock's own volatility and recent price moves. That is relevant for setting position limits, stop-loss levels, or review triggers. Institutional selling, often read as a negative signal, is a much weaker marker. Conversely, nothing in the results supports using the same flows to forecast average returns or market direction.

8.3 Limitations

The sample spans two years, a period in which the large-cap KOSPI rallied strongly and the KOSDAQ did not; the asymmetry results weaken in the second year. The skewness-based outcomes are noisy over 20 days. Standard errors for logit marginal effects are approximate. The investor-

type data aggregate heterogeneous traders into three groups; domestic institutions, in particular, combine pension funds, insurers, mutual funds, and proprietary desks with very different motives. Finally, the hypotheses originated in exploratory work on the same data, so the in-sample tests are confirmatory only in the limited sense that the specification was fixed in advance; the forward test is the real check.

9 Analysis plan: deviations and disclosures

The analysis plan was written on 27 October 2026 and stored with the code; it was not registered with an external registry. The following deviations and disclosures apply.

1. During construction of the variables, and before the confirmatory regressions, a diagnostic printout displayed the raw (uncontrolled) frequencies of the tail event by extreme-flow signal. The printout was removed and recorded in the plan's change log; no hypothesis, variable, or test was changed.
2. An error in the first construction of beta (a missing value in the first index return propagated through rolling sums) was found and fixed before any outcome-based analysis.
3. The wording of H8 did not anticipate significantly negative after-cost alphas; the formal outcome is reported as planned (Section 5).
4. The interpretations of the cumulative-flow, -10% threshold, and 60-day results (Section 7) were not anticipated in the plan and are labeled exploratory.

10 Conclusion

Korean investor-type flows, published daily for every stock, are a poor guide to average returns but a useful guide to the left tail. Heavy one-day foreign selling raises the probability of a large one-month underperformance by about a third, after controlling for the usual predictors of volatility and crashes, in both exchanges and in every random half of the cross-section. Selling by foreign investors is riskier than their buying, and buying by individuals is riskier than their selling, but these asymmetries weakened in the most recent year. A forward test on post-October 2026 data, scheduled in the analysis plan, will show whether the results survive outside the sample in which they were found.

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A Variable definitions

Net-buying intensity $x^j_{i,d}$. Net purchase value of investor group j (foreign; institutions; individuals) in stock i on day d, divided by the day-d closing market capitalization. Zero if the group did not trade.

Extreme signals S^j, B^j. Bottom and top deciles of $x^j_{i,d}$ within the stock's exchange on day d. Cumulative variants replace $N^j_{i,d}$ with the sum over the last $k = 5$ or 20 days. The heavy-trading variant sorts stocks into size terciles, then into terciles of gross trading intensity (buy plus sell value over market cap) of group j, and flags the bottom and top 30% of x^j within the highest-activity cell.

Holding return $R^h_{i,d}$. Compounded daily KRX return from the close of $d+1$ to the close of $d+1+h$. Suspended days and days after delisting have zero return.

Market-adjusted return $e^h_{i,d}$. $R^h_{i,d}$ minus the equal-weighted mean of R^h over all universe stocks on the same exchange and signal day.

Tail event $Y_{i,d}$. One if $e^{20}_{i,d} \le -0.20$. Path-minimum variant: one if the lowest cumulative raw return over the 20 days after entry is -20% or worse.

NCSKEW, DUVOL. Computed from $w_{i,t} = \ln(1 + r_{i,t} - \hat{\beta}_{i,d} r_{m,t})$ over the holding window (minimum 15 of 20 or 45 of 60 observations): $\text{NCSKEW} = -n(n-1)^{3/2} \sum w^3 / [(n-1)(n-2)(\sum w^2)^{3/2}]$; $\text{DUVOL} = \ln[(n_u - 1) \sum_{\text{down}} w^2 / ((n_d - 1) \sum_{\text{up}} w^2)]$, where down (up) days have w below (above) the window mean.

Controls. Size: log market capitalization. Value: log of book value per share over price (missing if book value is non-positive). Momentum: cumulative return over trading days $d-251$ to $d-21$. Reversal: cumulative return over $d-20$ to $d-1$. Signal-day and entry-day returns: $r_{i,d}$, $r_{i,d+1}$. Volatility: standard deviation of daily returns over $d-19$ to d. Beta: slope of daily returns on the exchange index over 250 days (minimum 120). Amihud: mean of $|r|$ divided by trading value (KRW 100 million) over 20 days. Turnover: mean trading value over market cap over 20 days. Past NCSKEW: as above over $d-59$ to d.

B Additional tables

Table 7: Investor flows and conditional skewness

	Foreign			Institutional			Individual		
Dependent variable	β_S	β_B	$\beta_S - \beta_B$	β_S	β_B	$\beta_S - \beta_B$	β_S	β_B	$\beta_S - \beta_B$
NCSKEW, 20 days	0.005	0.002	0.003	-0.006	-0.018	0.012	0.012	0.010	0.001
	(0.42)	(0.15)	(0.37)	(-0.39)	(-1.61)	(1.07)	(0.91)	(0.83)	(0.12)
DUVOL, 20 days	-0.000	-0.005	0.004	-0.033	-0.038	0.005	0.008	0.003	0.005
	(-0.03)	(-0.33)	(0.44)	(-2.04)	(-3.07)	(0.47)	(0.57)	(0.24)	(0.41)
NCSKEW, 60 days†	0.029	0.012	0.017	0.051	0.027	0.024	0.006	0.032	-0.026
	(1.56)	(0.78)	(2.16)	(4.17)	(1.65)	(1.88)	(0.40)	(1.69)	(-2.63)
DUVOL, 60 days†	0.008	-0.003	0.010	-0.011	-0.016	0.006	-0.003	0.007	-0.010
	(0.48)	(-0.19)	(2.25)	(-0.97)	(-1.19)	(0.77)	(-0.23)	(0.48)	(-1.92)

Notes: Same daily cross-sectional regressions as Table 3 with the dependent variable replaced by the negative coefficient of skewness (NCSKEW) or the down-to-up volatility ratio (DUVOL) of firm-specific log returns over the holding window. Firm-specific returns are $\ln(1 + r_{i,t} - \hat{\beta}_{i,d} r_{m,t})$ with $\hat{\beta}$ estimated before the signal day. Newey–West t-statistics in parentheses. †426 signal days; seven non-overlapping windows.

Table 8: Constructed Korean factors

Factor	Days	Mean daily return (%)	Annualized volatility (%)
MKT	485	0.161	41.1
SMB	485	-0.151	18.8
HML	485	-0.026	26.5
MOM	485	0.071	25.8
STR	485	-0.003	27.3

Notes: Factors are built from the study universe (both markets pooled). MKT is the value-weighted universe return minus the daily CD (91-day) rate; SMB and HML use a 2×3 sort on size (median) and book-to-market (30/70), MOM a 2×3 sort on 12–1 month returns, both reformed at month-ends; STR is a daily-reformed 2×3 sort on the prior 20-day return (losers minus winners). Correlations of daily returns with KRX indices: our KOSPI index 1.0000, our KOSDAQ index 0.9996; SMB with the average of (KOSPI small − large) and (KOSDAQ small − large) size indices 0.66. The lower SMB correlation reflects pooled-median versus within-market rank breakpoints.